

CKI User Guide

Cell-state Kinetic Index — From Data to Biological Insight

A visual reference for practitioners

v1.0

1 Core Pipeline

Input → Classification → Divergence → Output

01 Input Data

Input

- Gene expression matrix (cells × genes)
- Housekeeping gene list (HK reference)
- Identity gene list (functional markers)

02 Classify Genes

Classify

- Housekeeping (HK), neutral reference
- Identity genes, functional markers
- Per-gene JS divergence (calibrated)

03 Compute Divergence

Compute

- HK → k_n (neutral divergence)
- Identity → k_f (functional)
- Bootstrap CI (uncertainty estimate)

04 Output ω

Output

- $\omega = k_f / k_n$
- Statistical significance (P value, FDR)
- Biological interpretation (selection pressure)

2 Interpret ω

$\omega = k_f / k_n$ — Interpretation Guide

$$\omega = k_f / k_n$$

$\omega < 0.5$

Convergent / Purifying selection — Transcriptome homogenization

$\omega \approx 0.5-1.5$

Neutral range — No directional selection

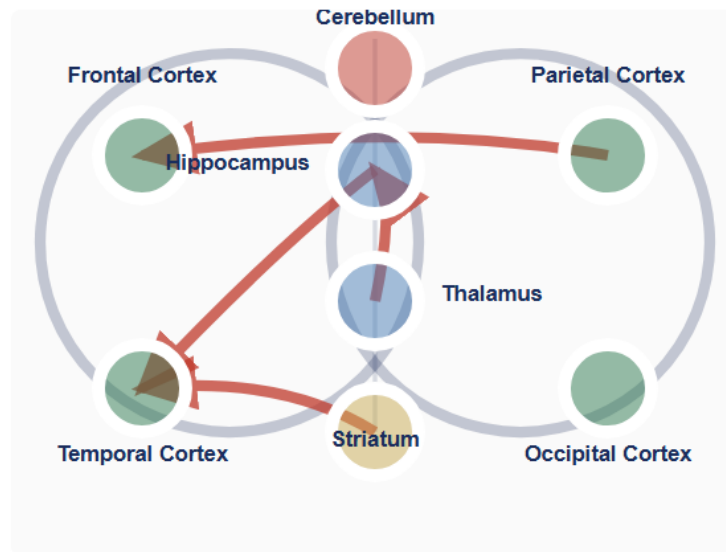
$\omega > 1.5$

Divergent / Positive selection — Transcriptome remodeling

k_f = Jensen-Shannon divergence of identity genes
 k_n = Jensen-Shannon divergence of HK genes (neutral ref.)
 $\omega < 1$ = Transcriptome convergence (purifying)
 $\omega > 1$ = Transcriptome divergence (positive selection)

3 Case Study: Human Brain Cell Migration

Siletti et al. 888K nuclei, ~100 regions | OPC migration analysis



Cell Type	ω	P value	Evidence
OPC	1.19	3.2×10^{-4}	Vascular-guided migration
Astrocyte	0.87	0.012	Regional specification
Microglia	0.72	0.045	Immune homing

Frontal → Temporal migration
 Consistent with vascular-guided OPC migration
 (Tsai 2016, Akay 2022)

213/31,740 significant migration candidates (0.67%)

★ Strongest signal: OPC ($\omega=1.19$)

⚡ Quick Start — 3 lines to run CKI

```
$ pip install cki
>>> import cki
>>> results = cki.compute(adata, cell_type="OPC", region="Frontal", region2="Temporal")
```

```
/// result = cki.compute(adata, nk_genes, identity_genes)
```

github.com/zhanglikun/CKI-cell-type-identification | `pip install cki` | MIT License